

Universal Linear Algebra Performance Audit (FAST) (FAST)

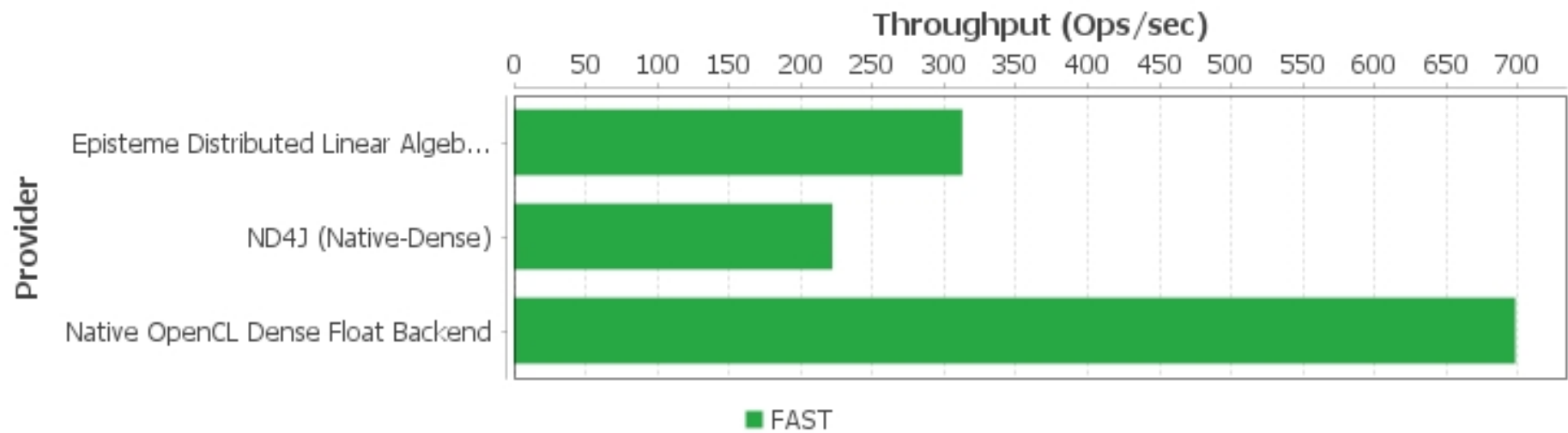
Audit Context:

Precision: fast
Environment: GCP
Matrix Size: 5x5
Reconstructed: True (Live from JSON)

Provider	Operation	Domain	Result Summary
DistributedLinearAlgebraProvider	Episteme Distributed Linear Algebra Provider (LocalDistributedContext)	Linear Algebra Performance	39 operations (avg 7,980 ms)
NativeND4JLinearAlgebraBackend	ND4J (Native-Dense)	Linear Algebra Performance	17 operations (avg 3,847 ms)
NativeOpenCLDenseLinearAlgebraFloatBackend	Native OpenCL Dense Float Backend	Linear Algebra Performance	23 operations (avg 6,977 ms)
NativeOpenCLSparseLinearAlgebraFloatBackend	Native OpenCL Sparse Float Backend	Linear Algebra Performance	8 operations (avg 2,747 ms)
GRPCLinearAlgebraBackend	Episteme gRPC Remote (localhost:50051)	Linear Algebra Performance	No successful operations

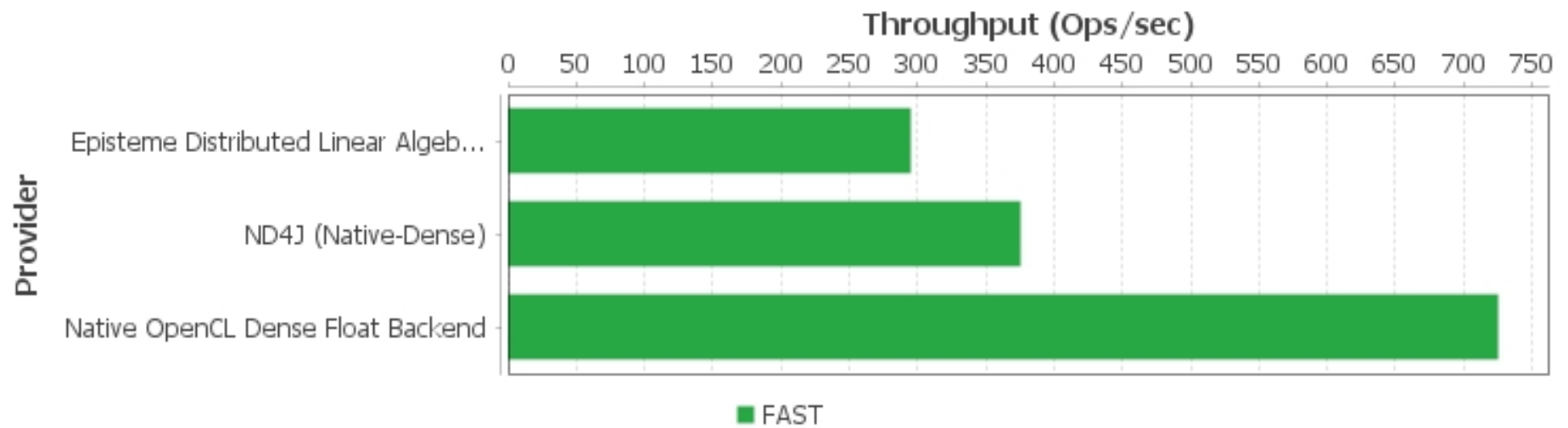
Operation Comparison: R:Add (FAST)

R:Add Performance Comparison



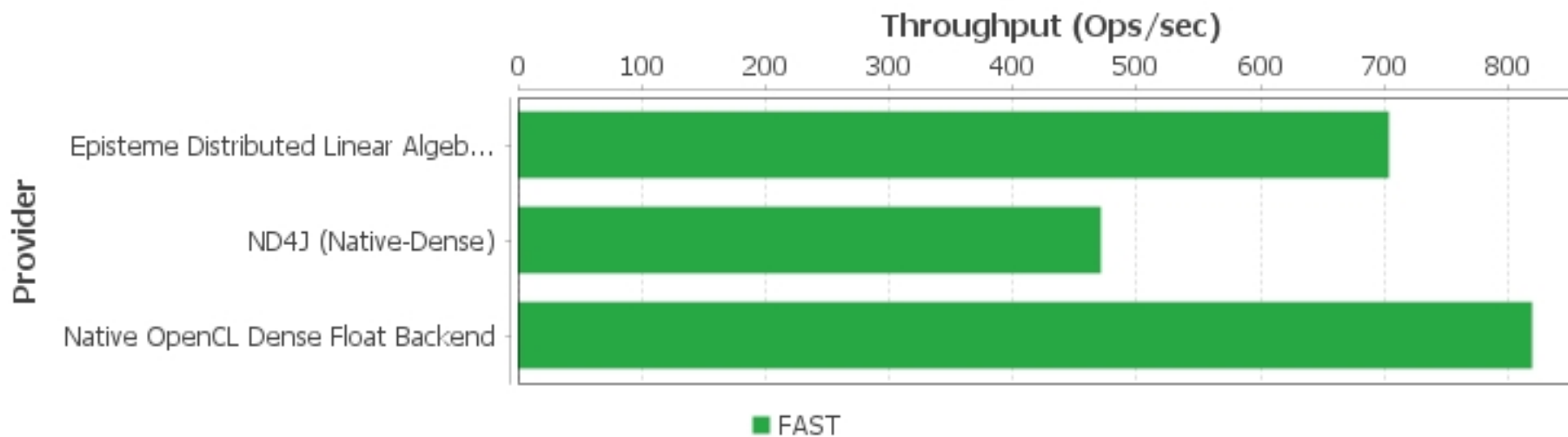
Operation Comparison: R:Sub (FAST)

R:Sub Performance Comparison



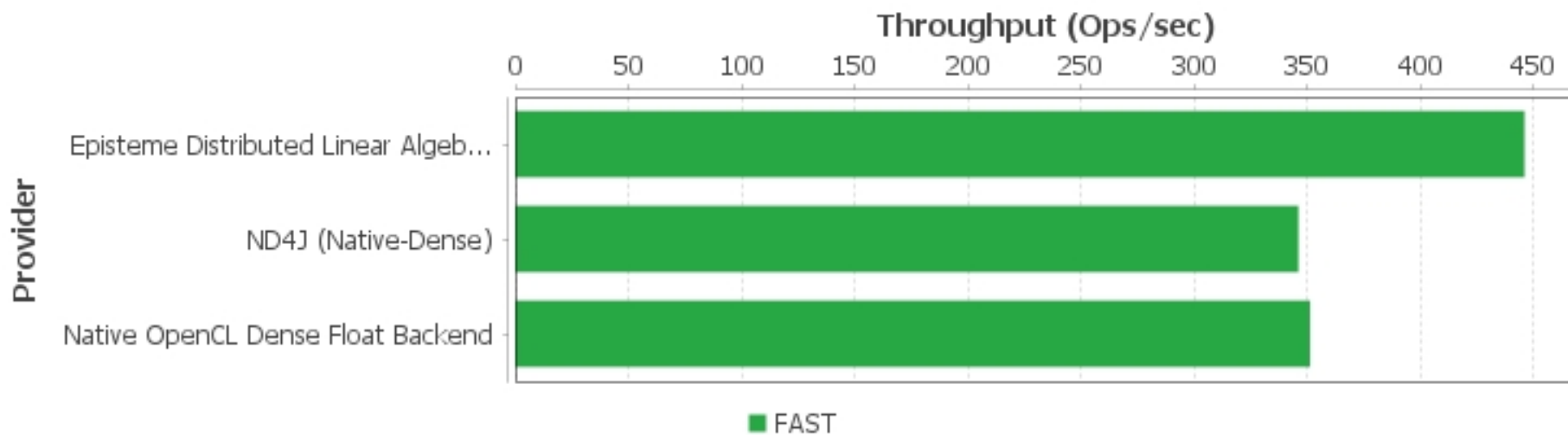
Operation Comparison: R:Scale (FAST)

R:Scale Performance Comparison



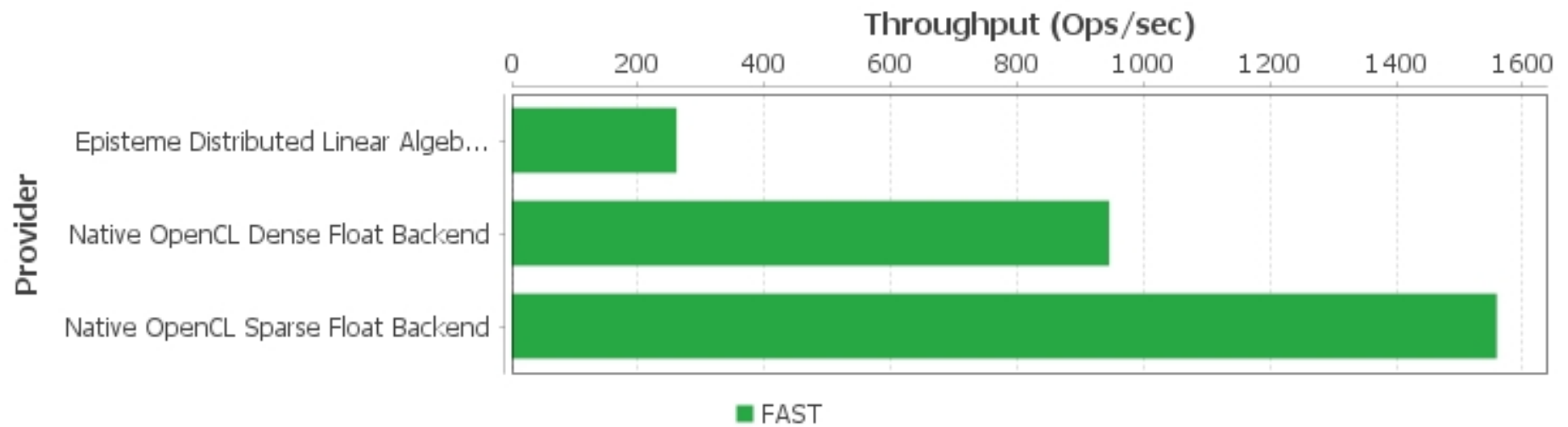
Operation Comparison: R:Mul (FAST)

R:Mul Performance Comparison



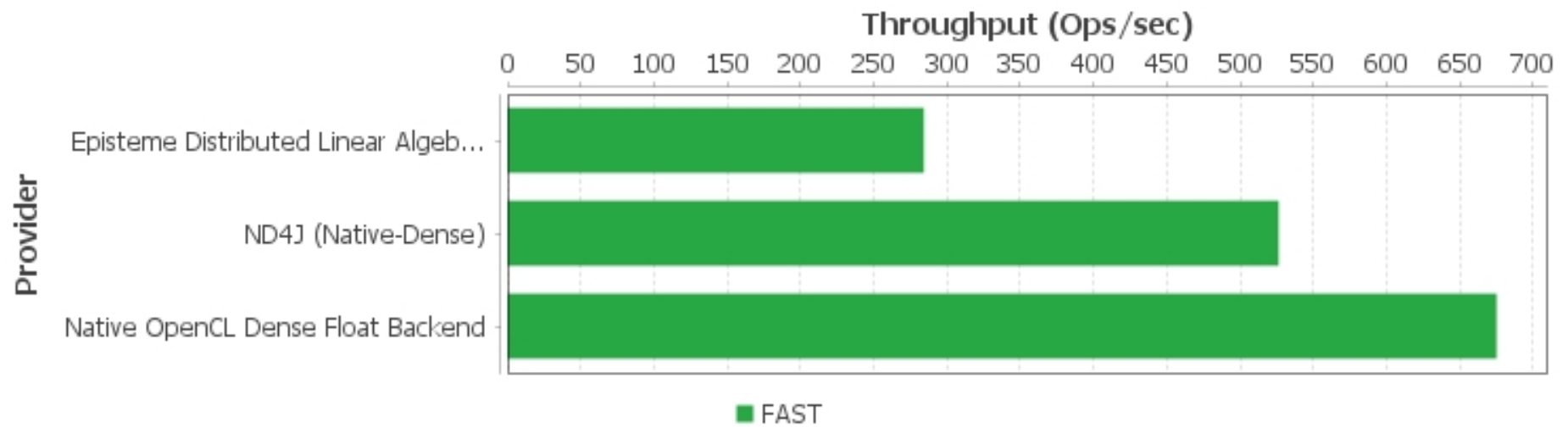
Operation Comparison: R:MatVec (FAST)

R:MatVec Performance Comparison

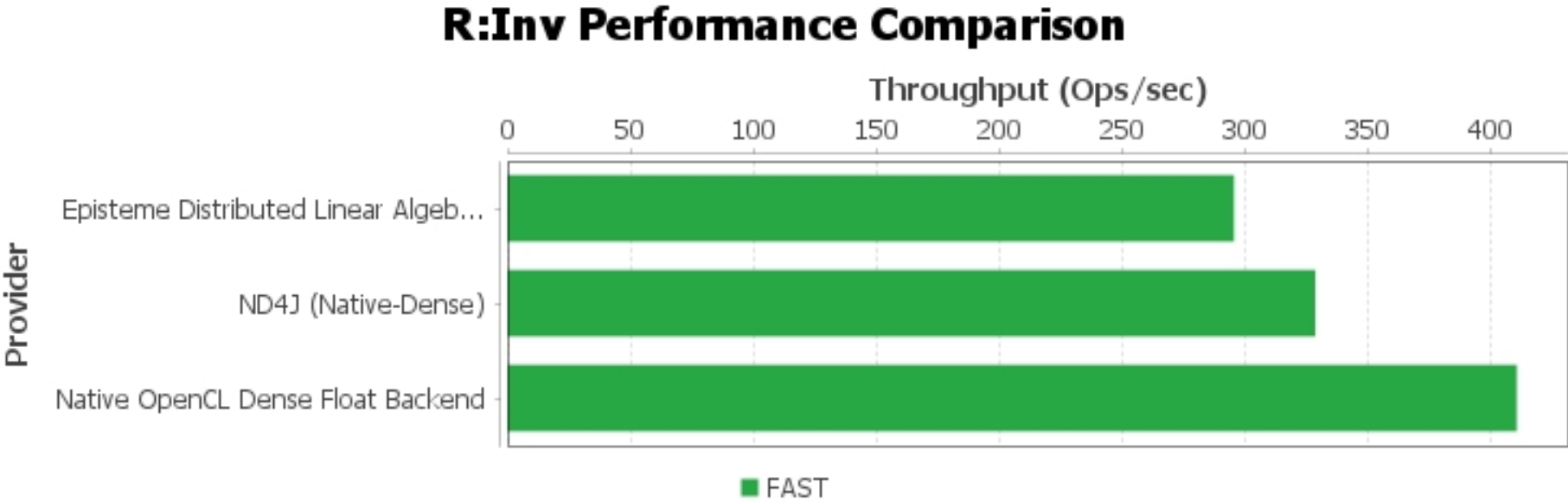


Operation Comparison: R:Trans (FAST)

R:Trans Performance Comparison

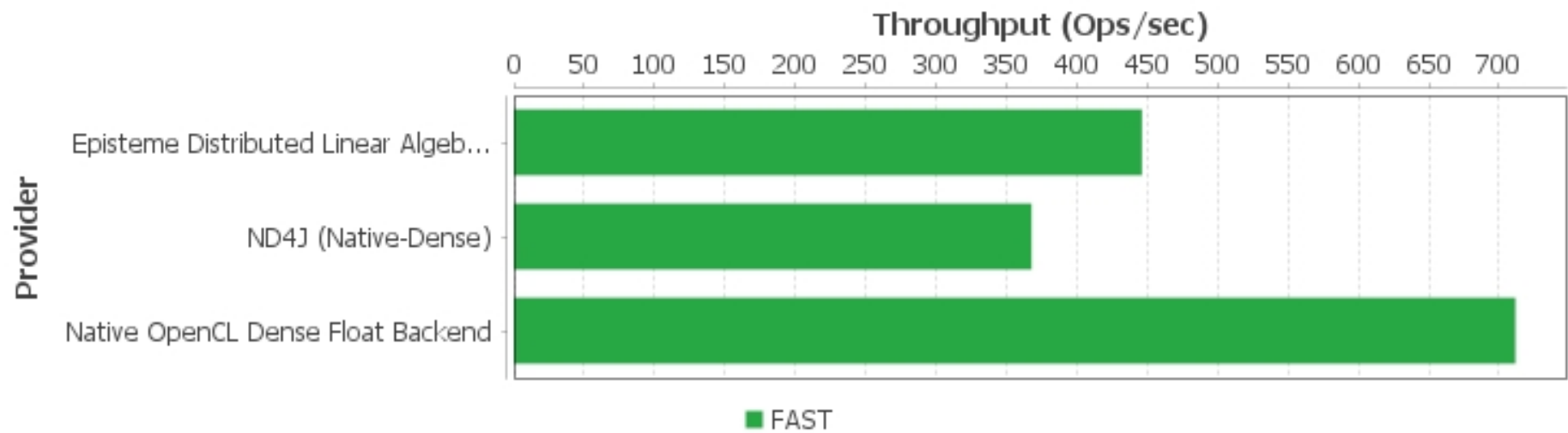


Operation Comparison: R:Inv (FAST)

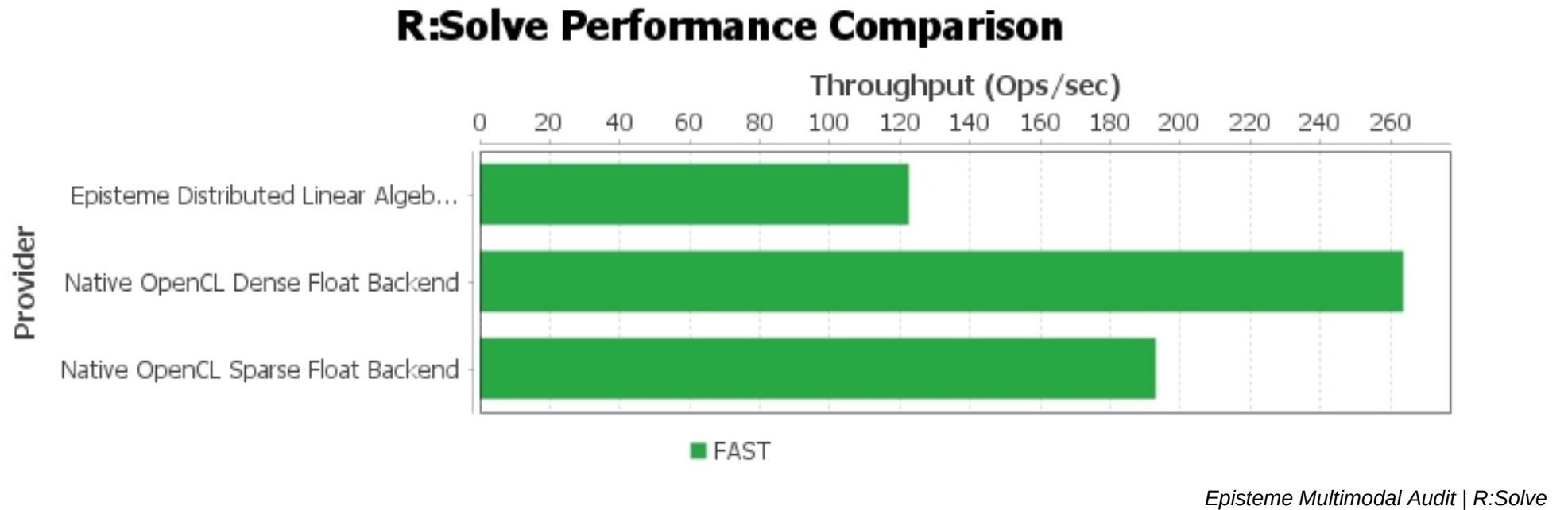


Operation Comparison: R:Det (FAST)

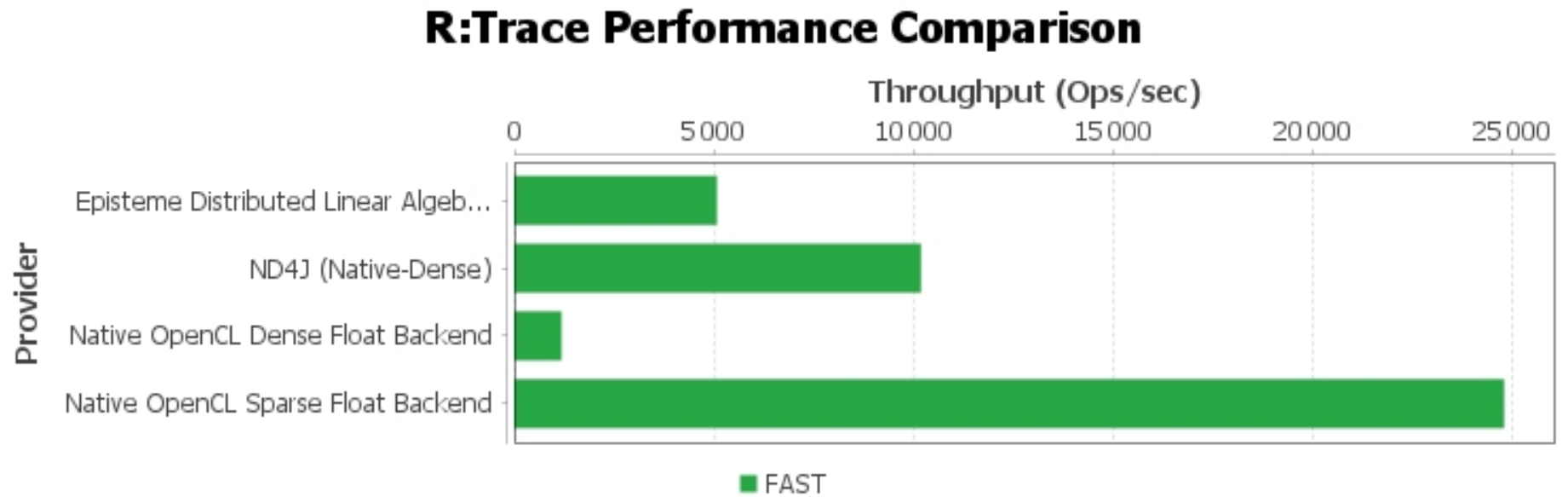
R:Det Performance Comparison



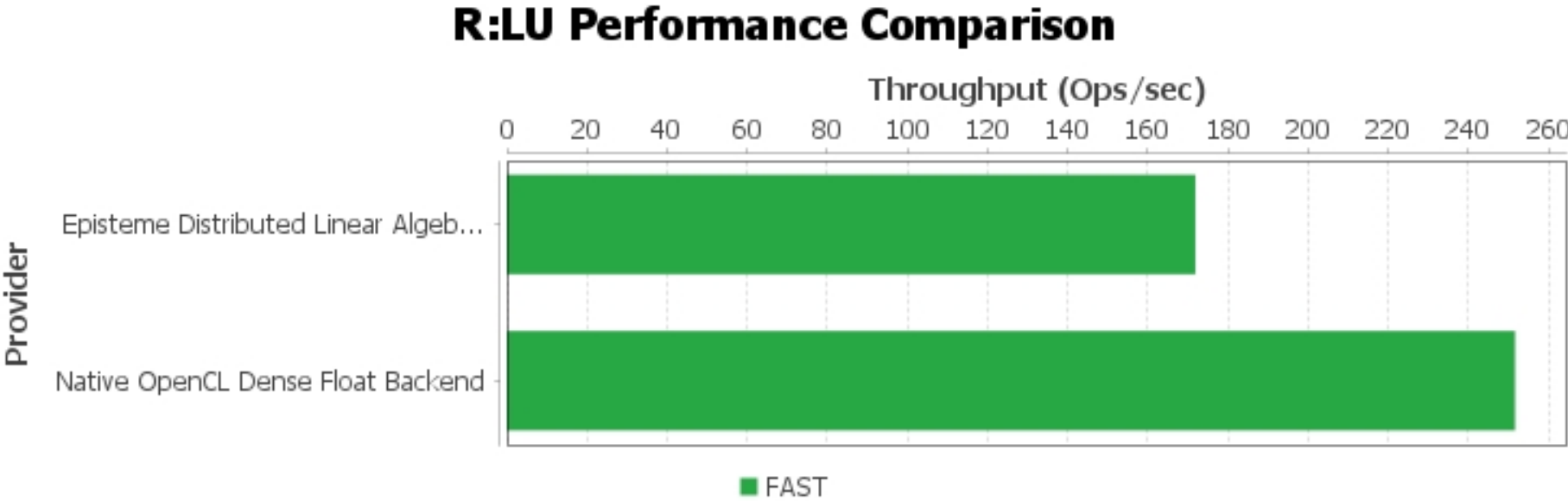
Operation Comparison: R:Solve (FAST)



Operation Comparison: R:Trace (FAST)

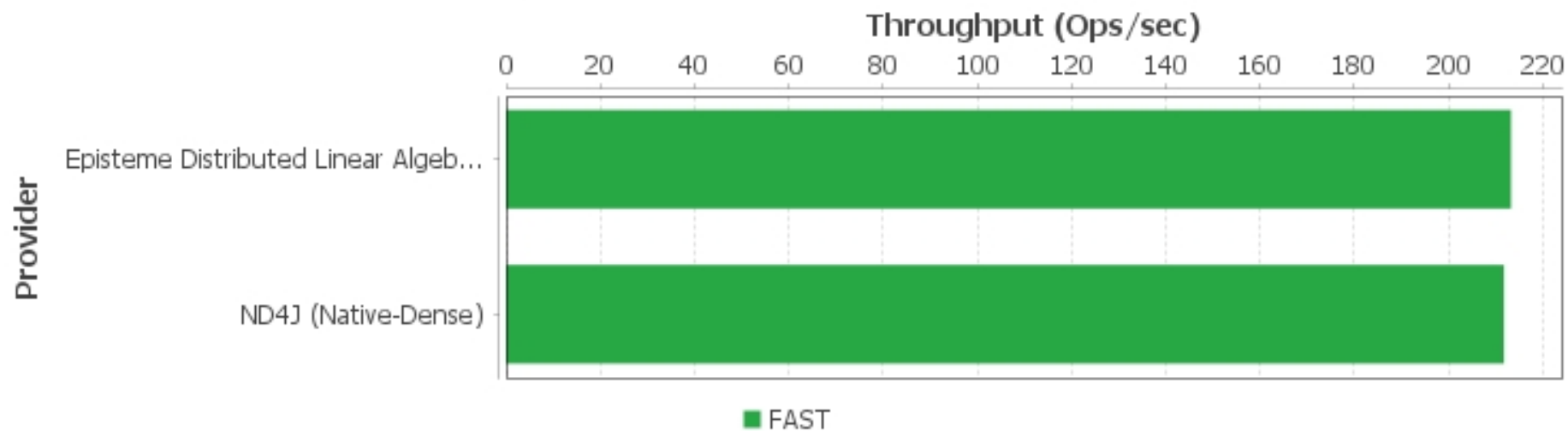


Operation Comparison: R:LU (FAST)



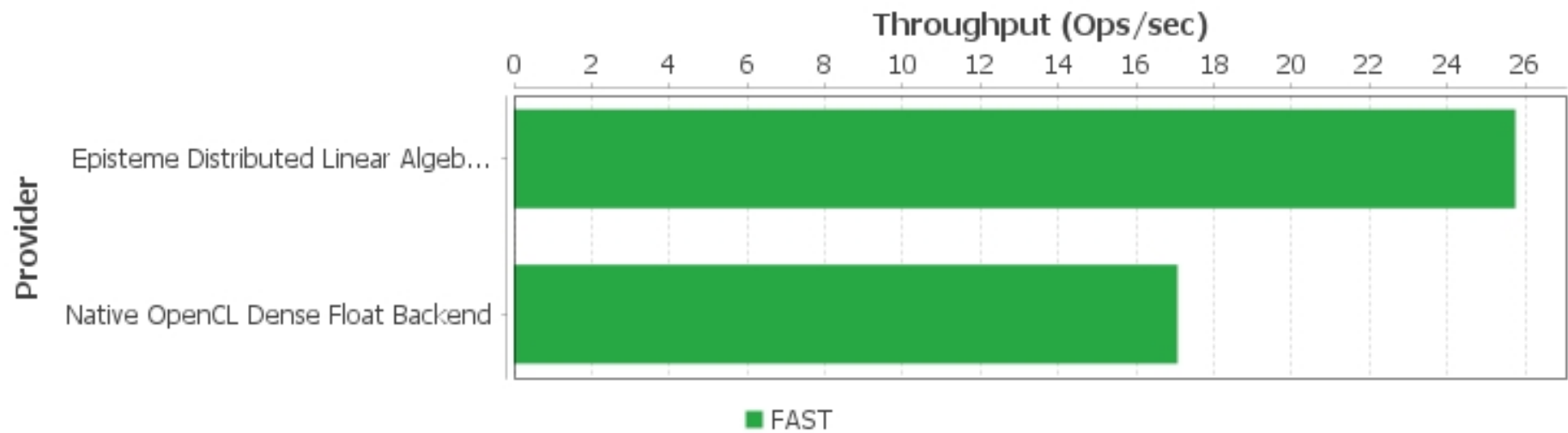
Operation Comparison: R:QR (FAST)

R:QR Performance Comparison

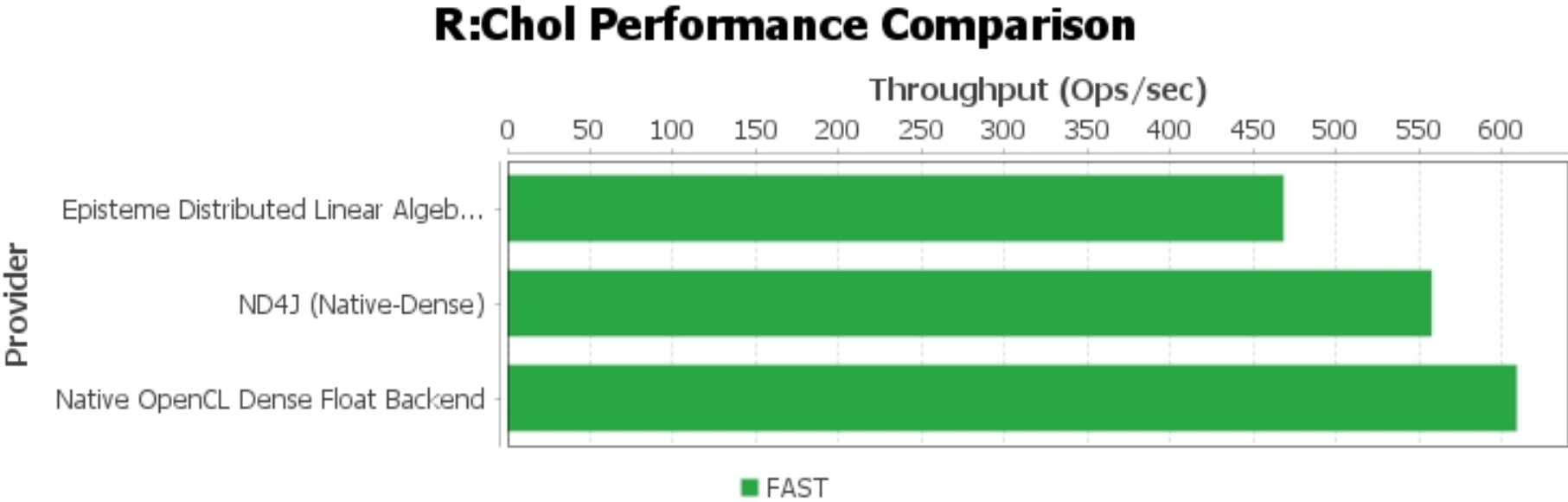


Operation Comparison: R:SVD (FAST)

R:SVD Performance Comparison

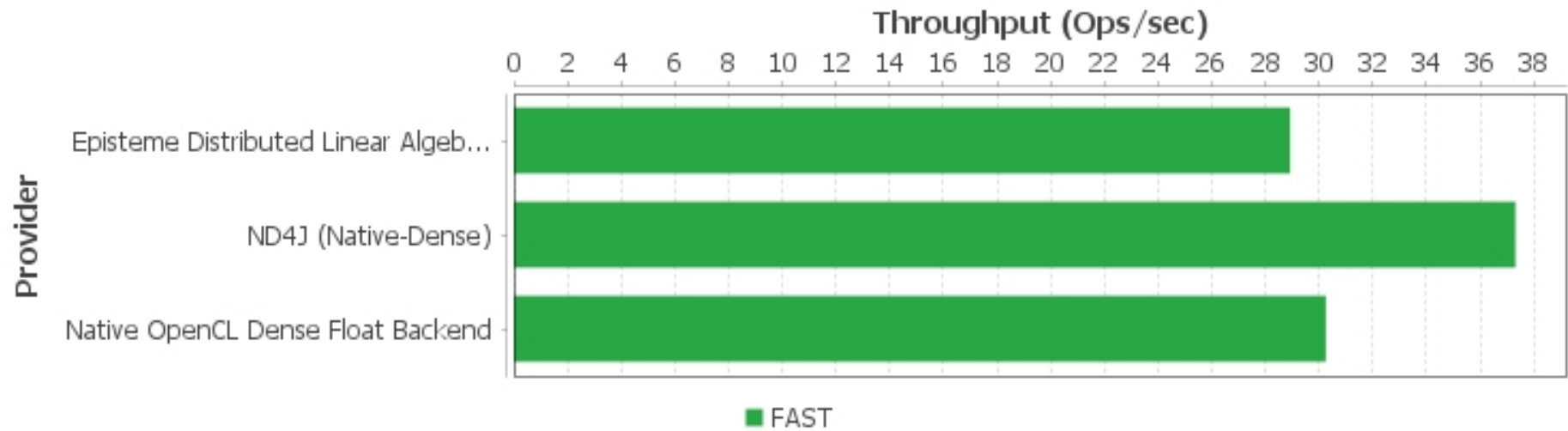


Operation Comparison: R:Chol (FAST)

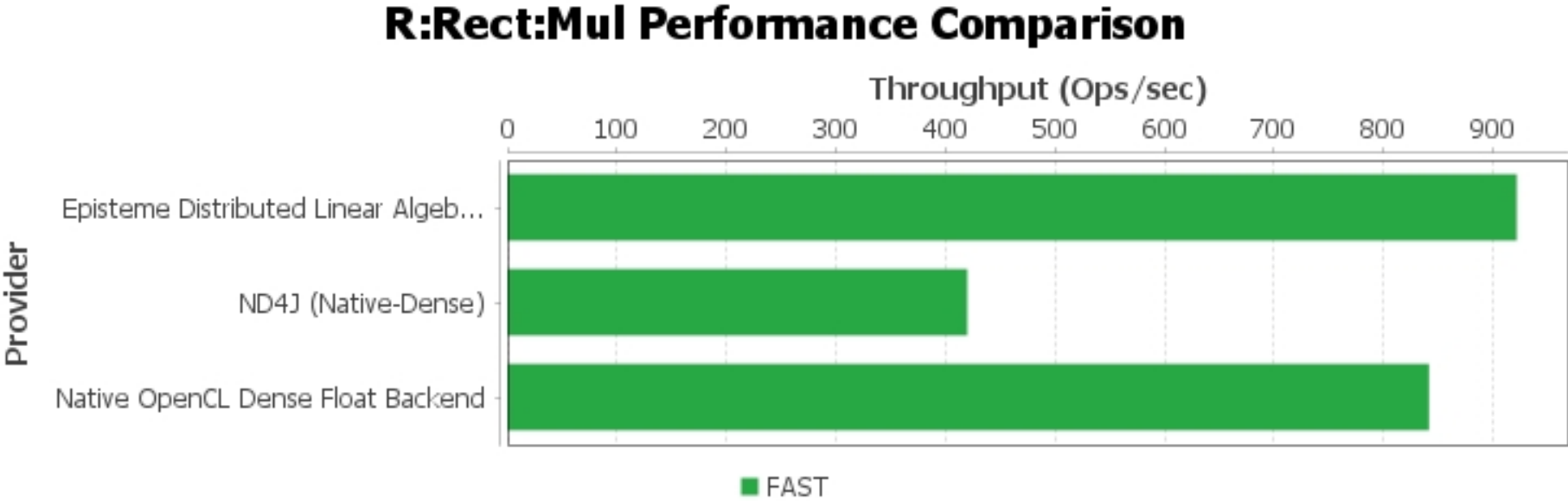


Operation Comparison: R:Eigen (FAST)

R:Eigen Performance Comparison

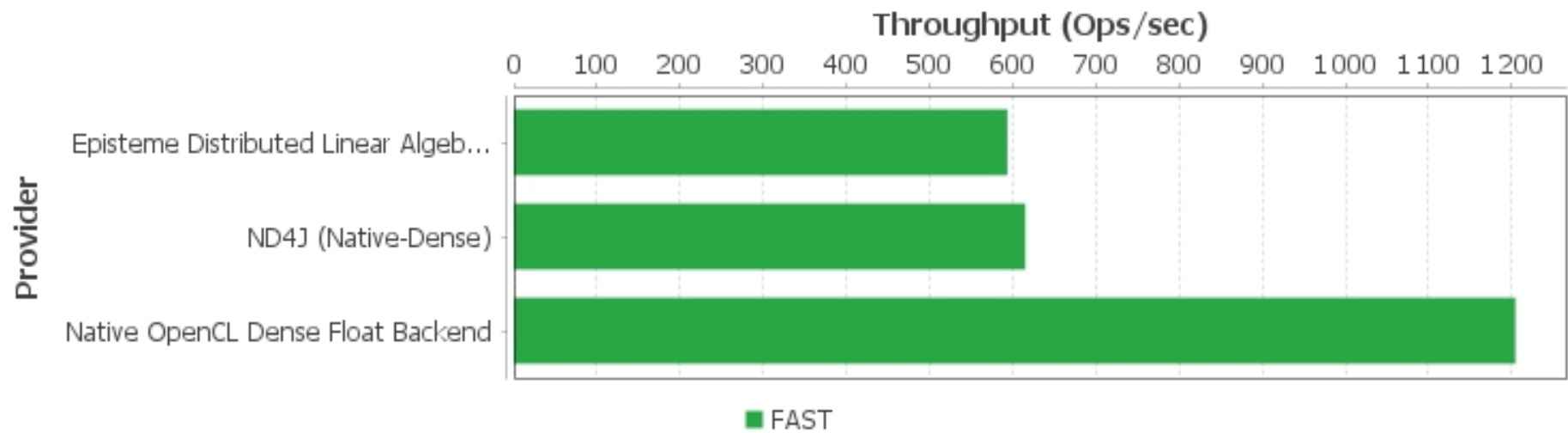


Operation Comparison: R:Rect:Mul (FAST)

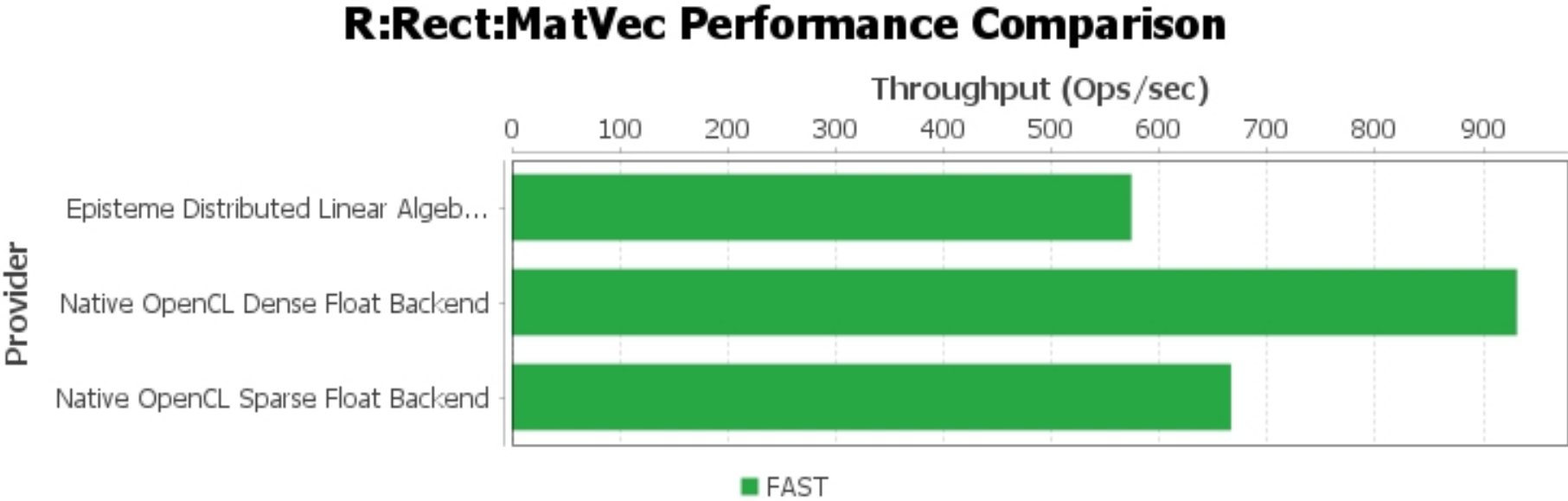


Operation Comparison: R:Rect:Trans (FAST)

R:Rect:Trans Performance Comparison

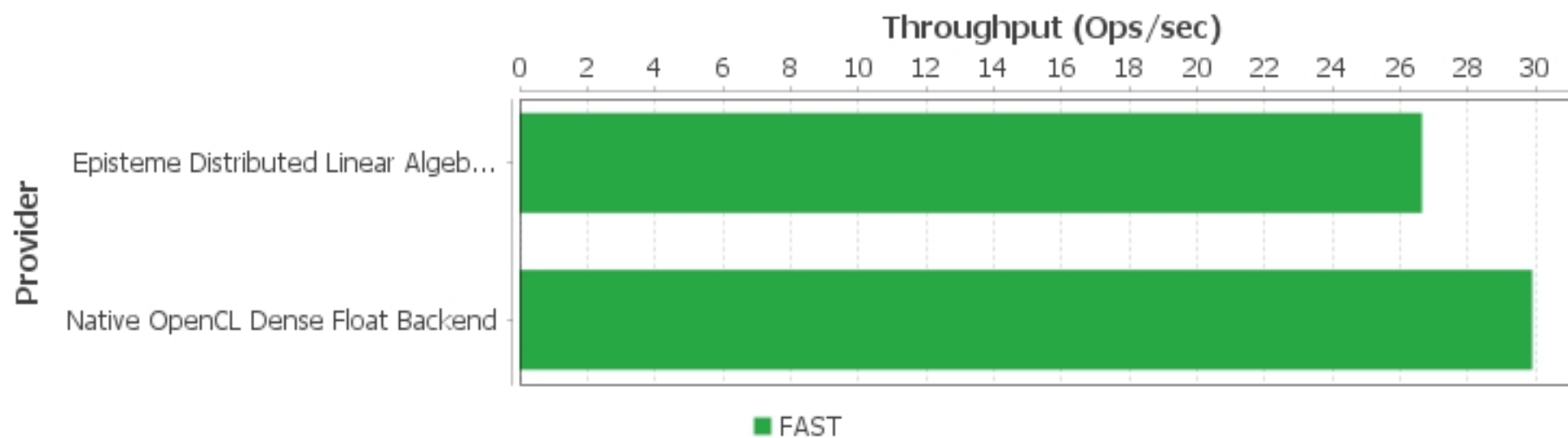


Operation Comparison: R:Rect:MatVec (FAST)



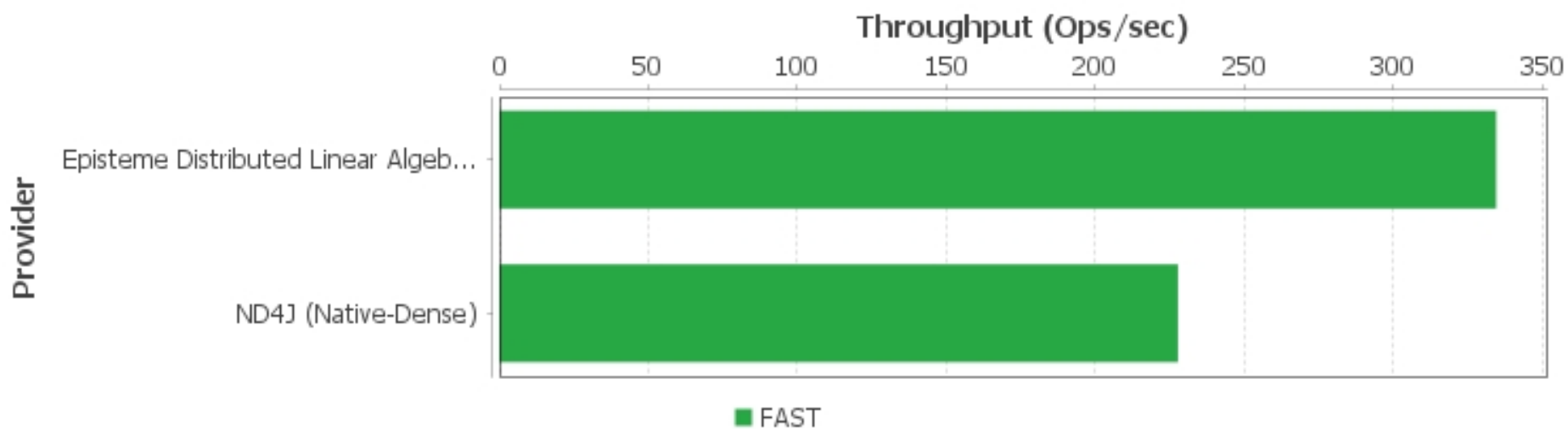
Operation Comparison: R:Rect:SVD (FAST)

R:Rect:SVD Performance Comparison

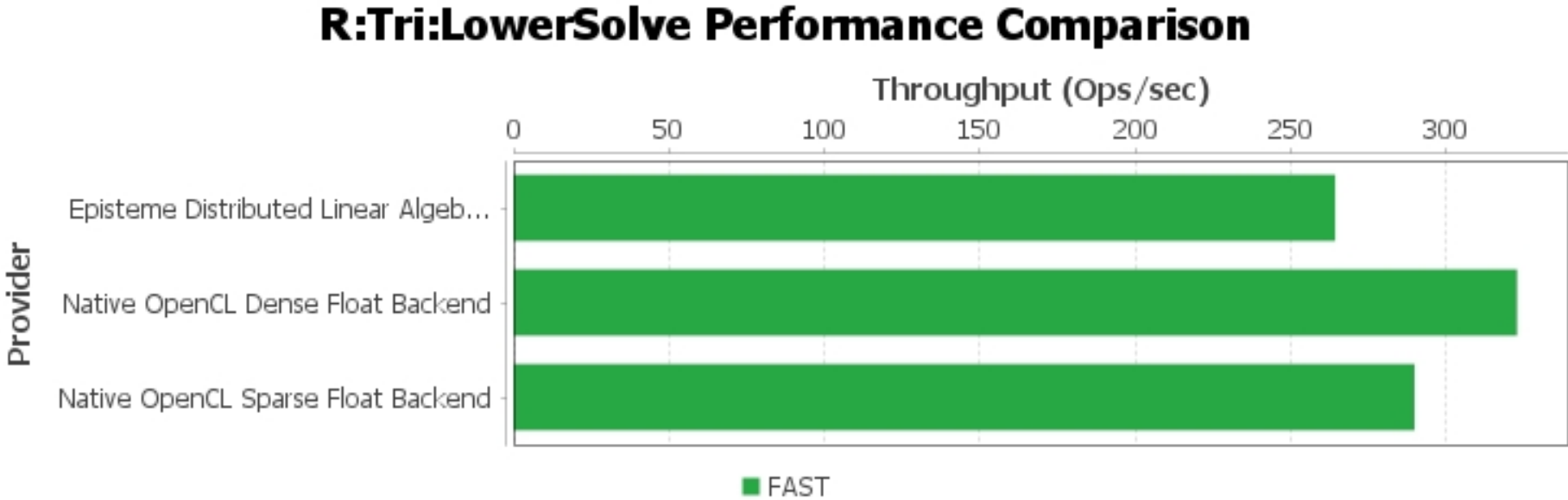


Operation Comparison: R:Rect:QR (FAST)

R:Rect:QR Performance Comparison



Operation Comparison: R:Tri:LowerSolve (FAST)



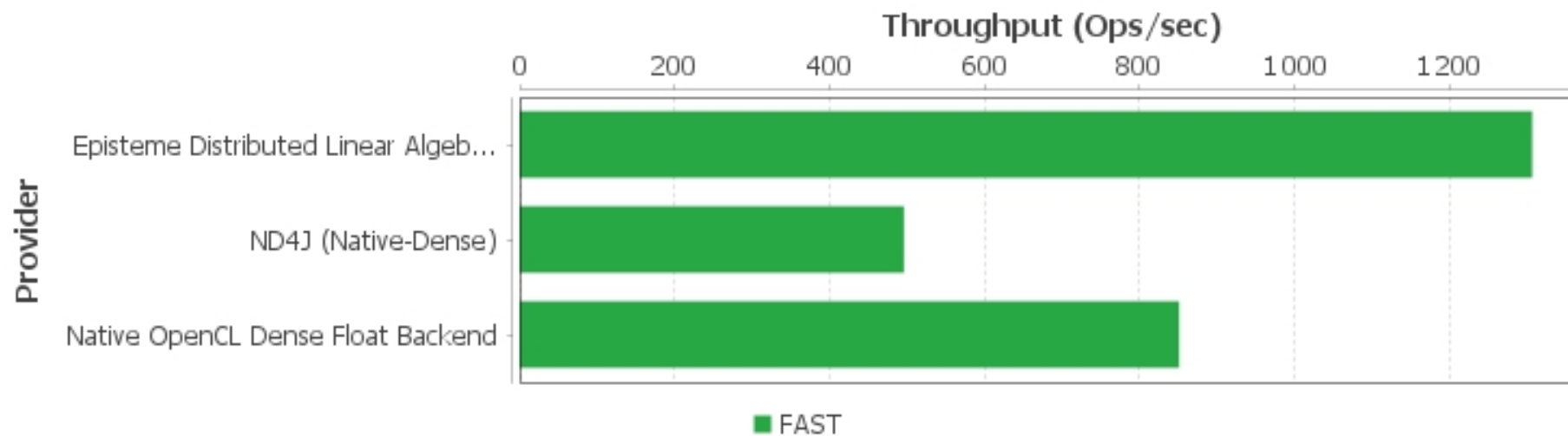
Operation Comparison: R:Tri:UpperSolve (FAST)

R:Tri:UpperSolve Performance Comparison



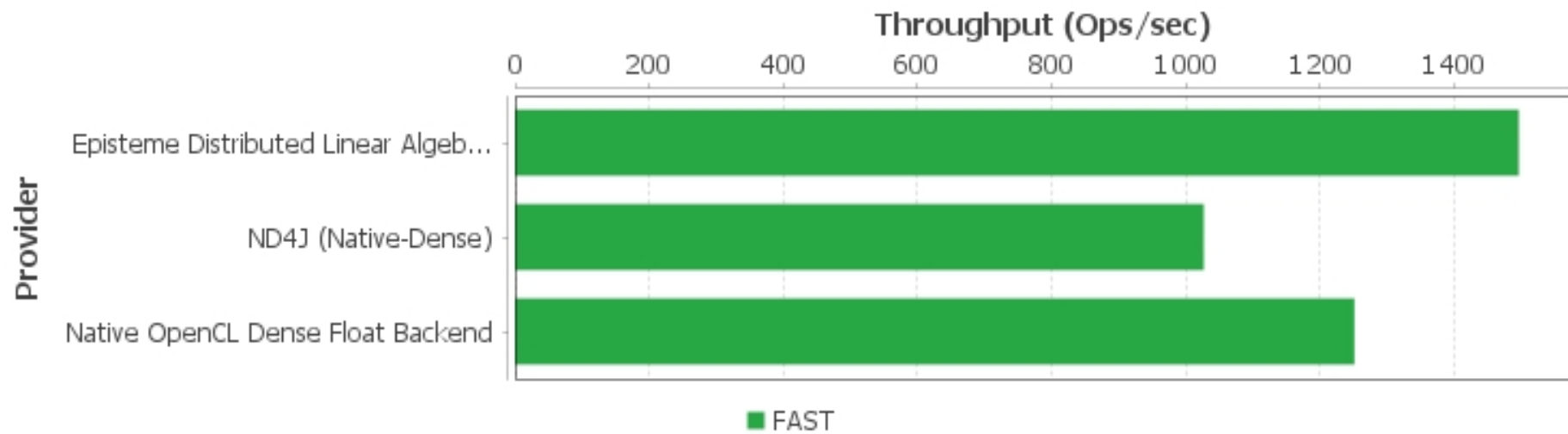
Operation Comparison: R:Tri:Mul (FAST)

R:Tri:Mul Performance Comparison



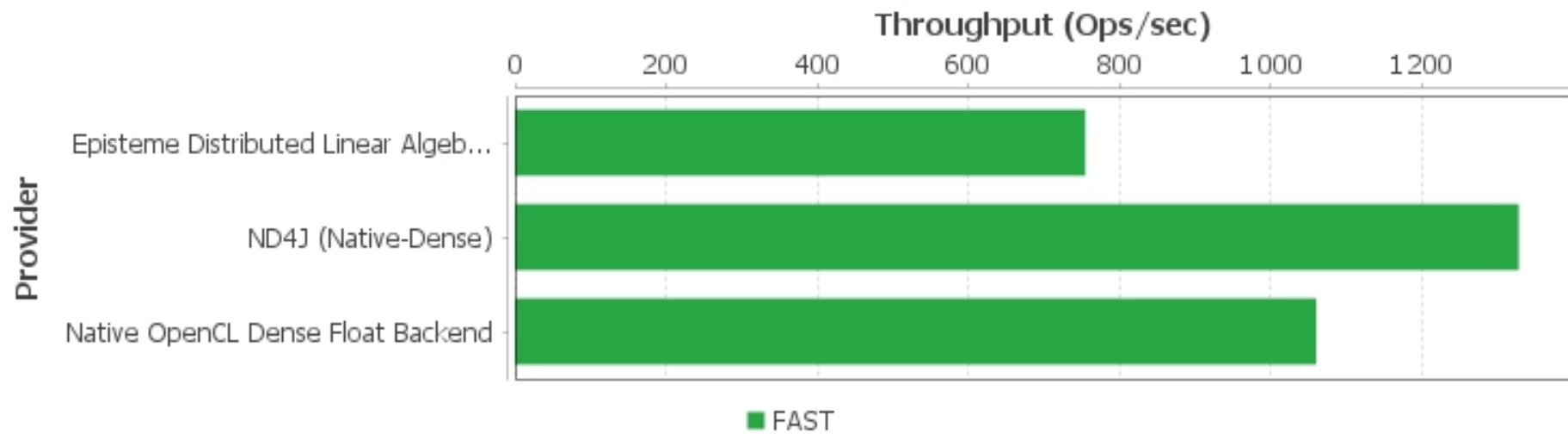
Operation Comparison: R:Vec:Dot (FAST)

R:Vec:Dot Performance Comparison



Operation Comparison: R:Vec:Norm (FAST)

R:Vec:Norm Performance Comparison



Operation Comparison: R:Vec:Normalize (FAST)

Operation Comparison: R:Vec:Cross (FAST)

Episteme Multimodal Audit | R:Vec:Cross

Operation Comparison: R:Vec:Angle (FAST)

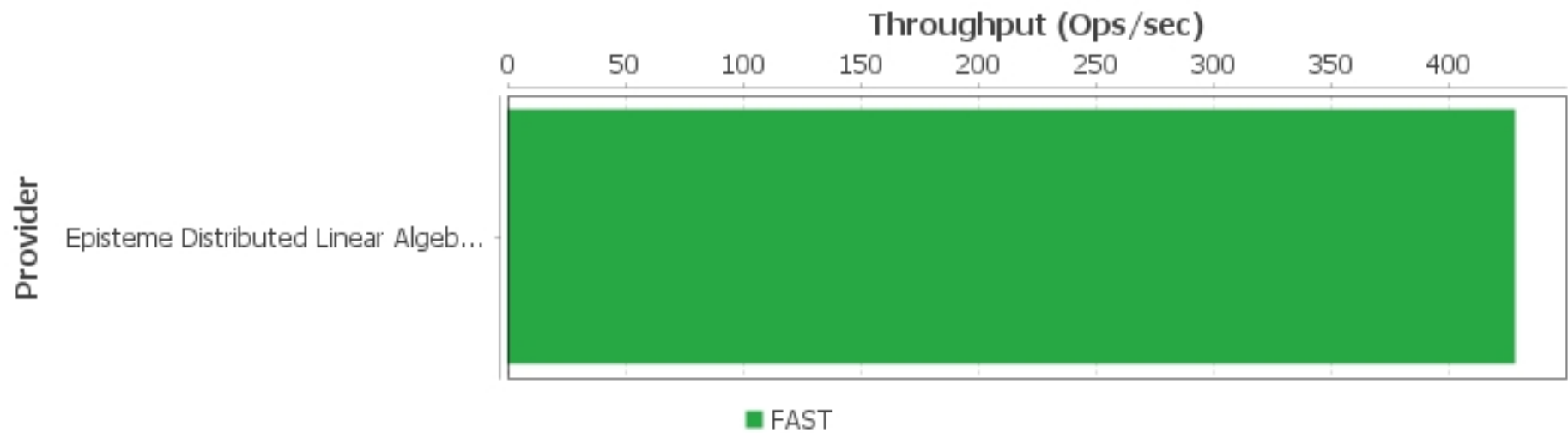
Episteme Multimodal Audit | R:Vec:Angle

Operation Comparison: R:Vec:Proj (FAST)

Episteme Multimodal Audit | R:Vec:Proj

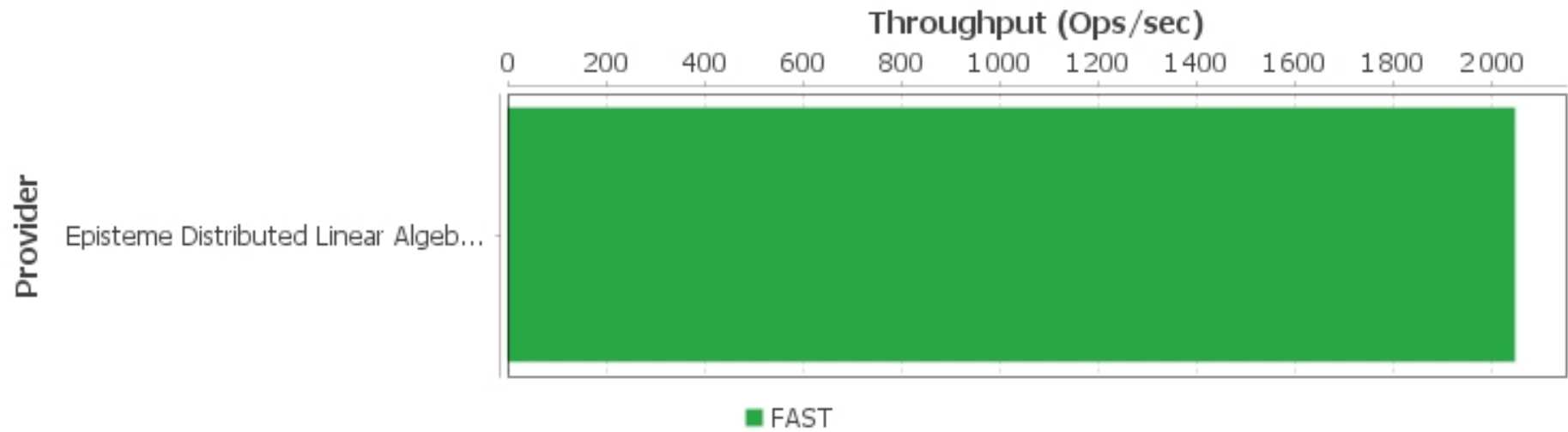
Operation Comparison: R:Func:Exp (FAST)

R:Func:Exp Performance Comparison

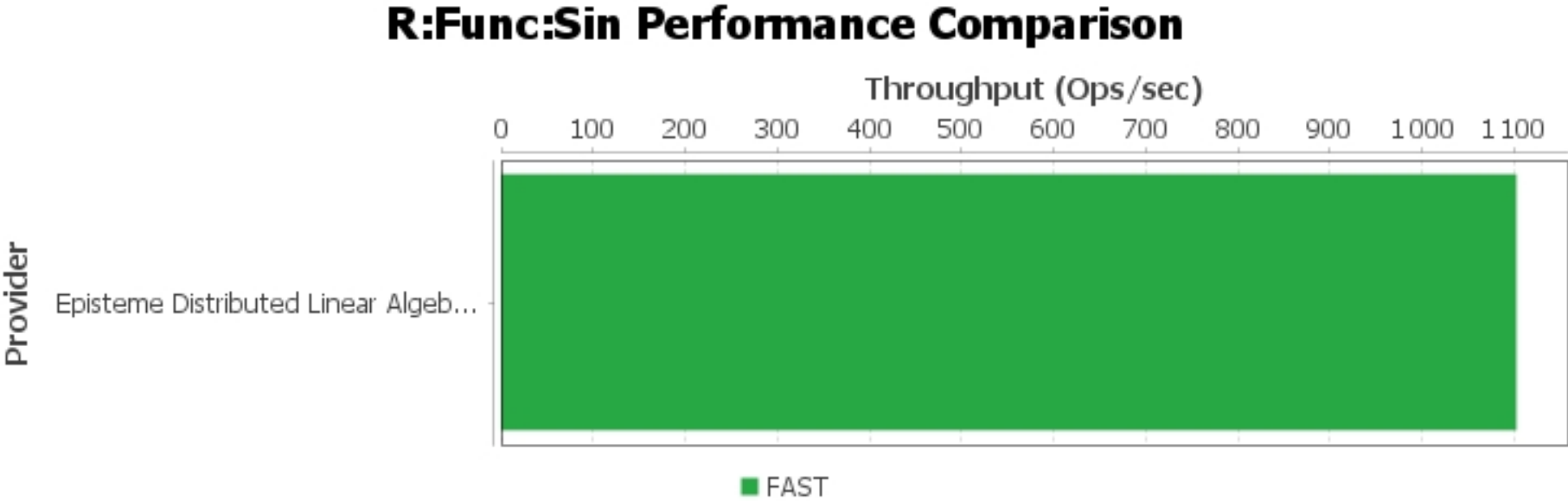


Operation Comparison: R:Func:Log (FAST)

R:Func:Log Performance Comparison

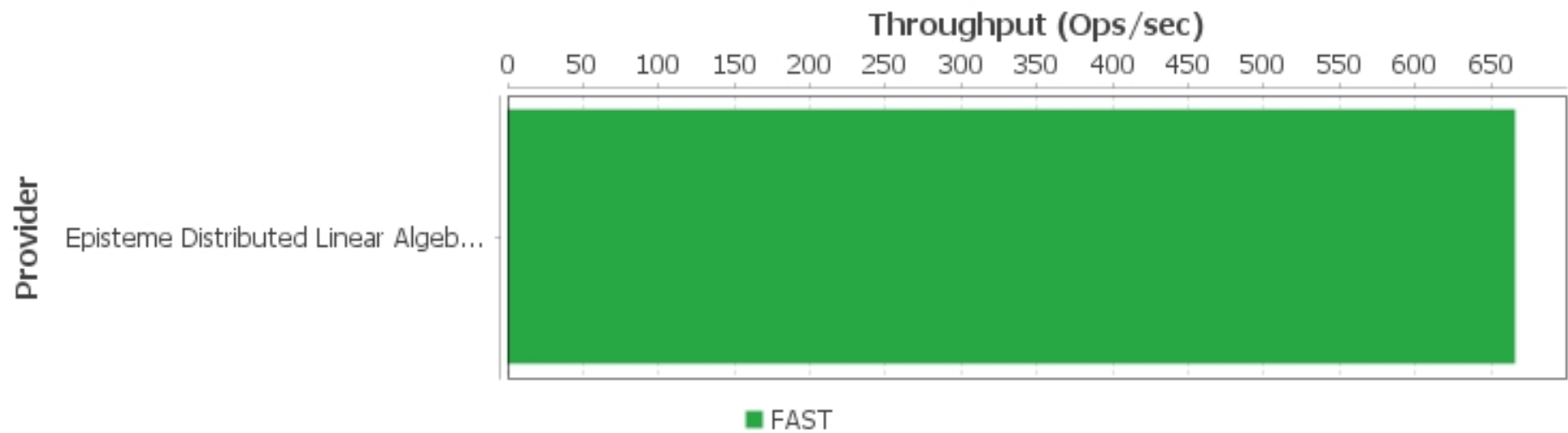


Operation Comparison: R:Func:Sin (FAST)



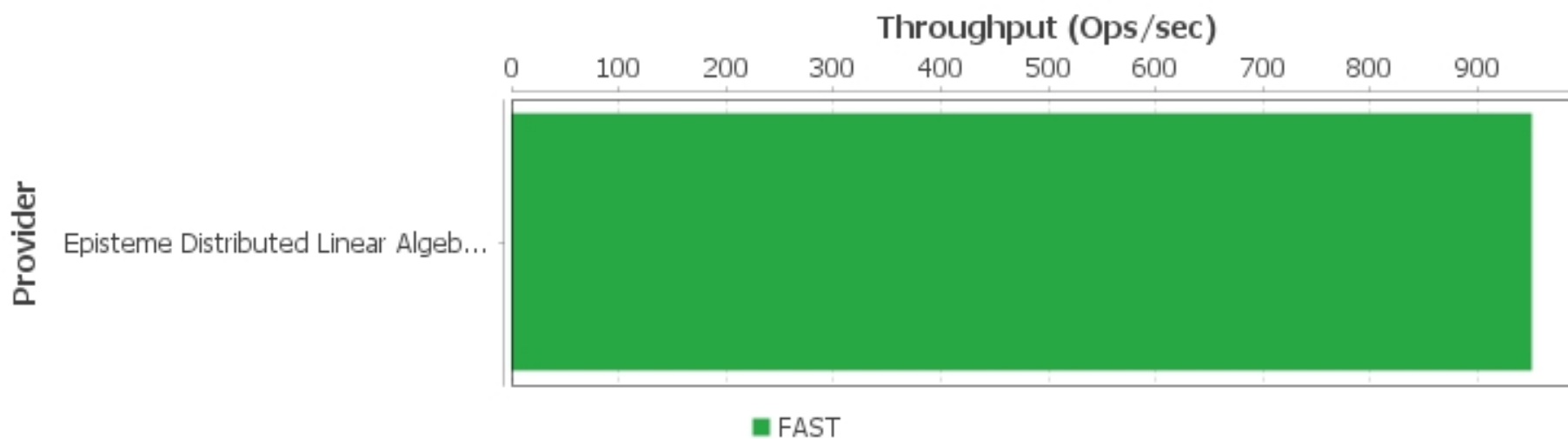
Operation Comparison: R:Func:Cos (FAST)

R:Func:Cos Performance Comparison



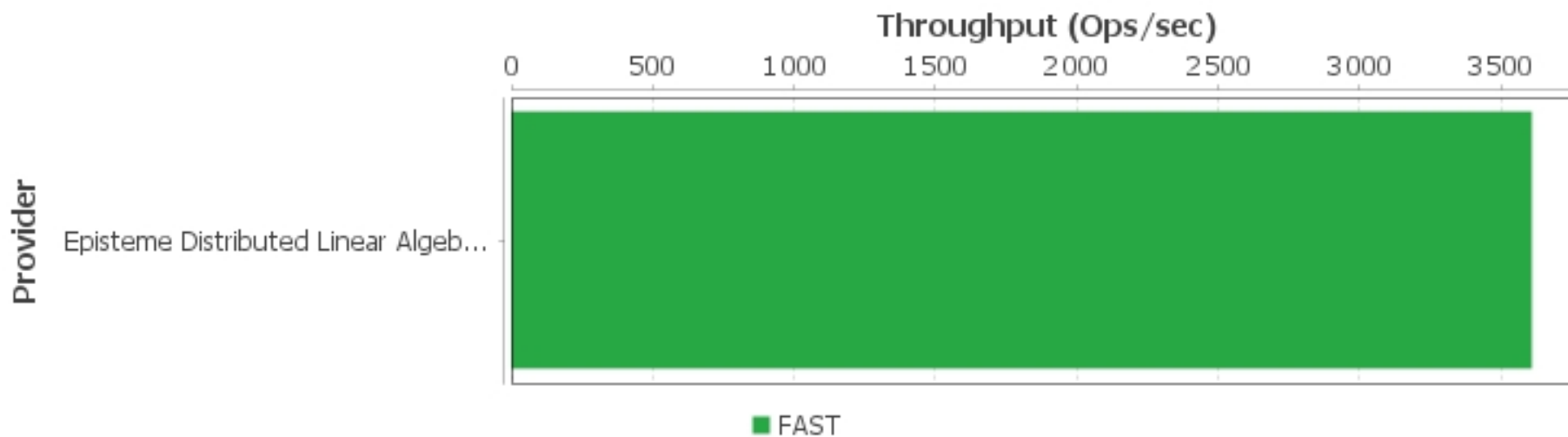
Operation Comparison: R:Func:Tan (FAST)

R:Func:Tan Performance Comparison

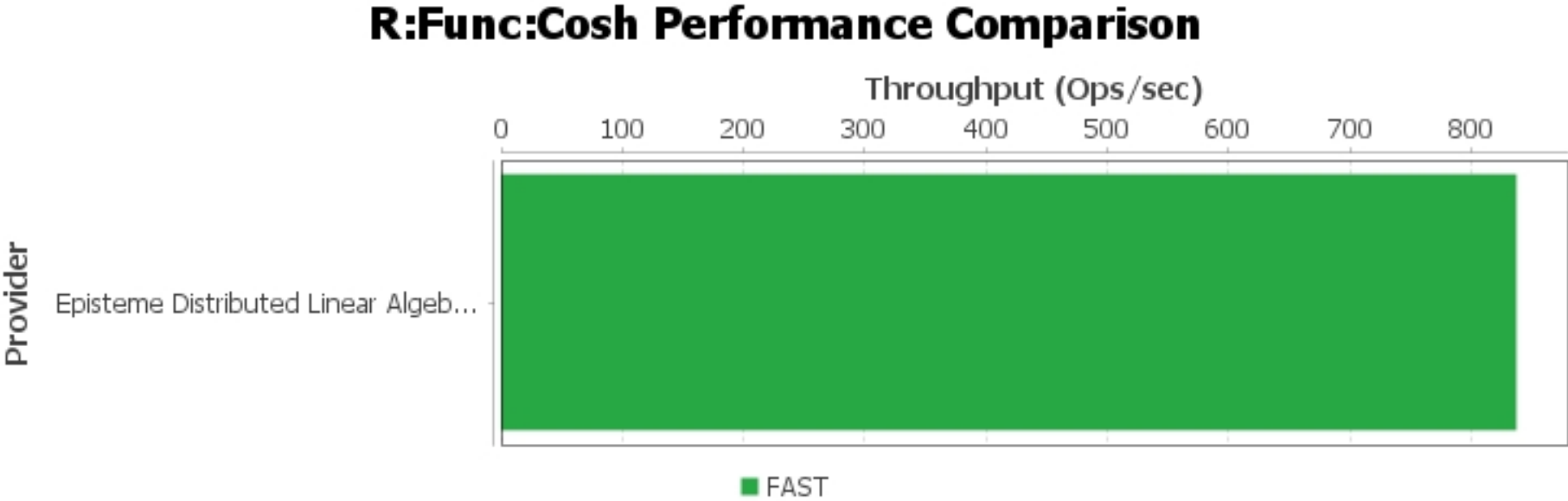


Operation Comparison: R:Func:Sinh (FAST)

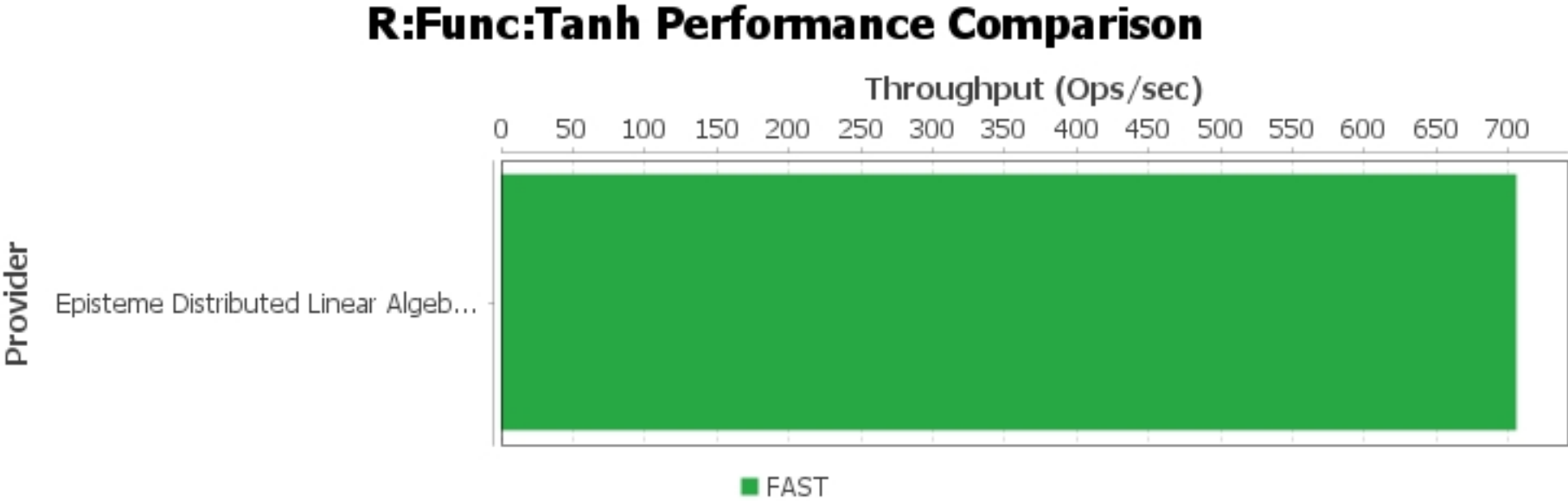
R:Func:Sinh Performance Comparison



Operation Comparison: R:Func:Cosh (FAST)

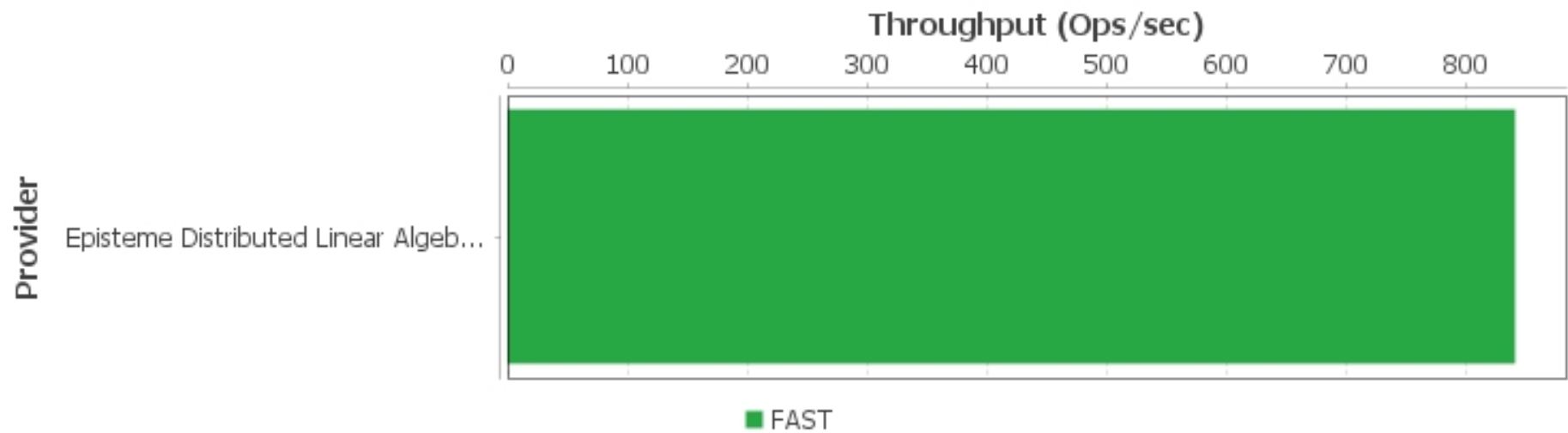


Operation Comparison: R:Func:Tanh (FAST)



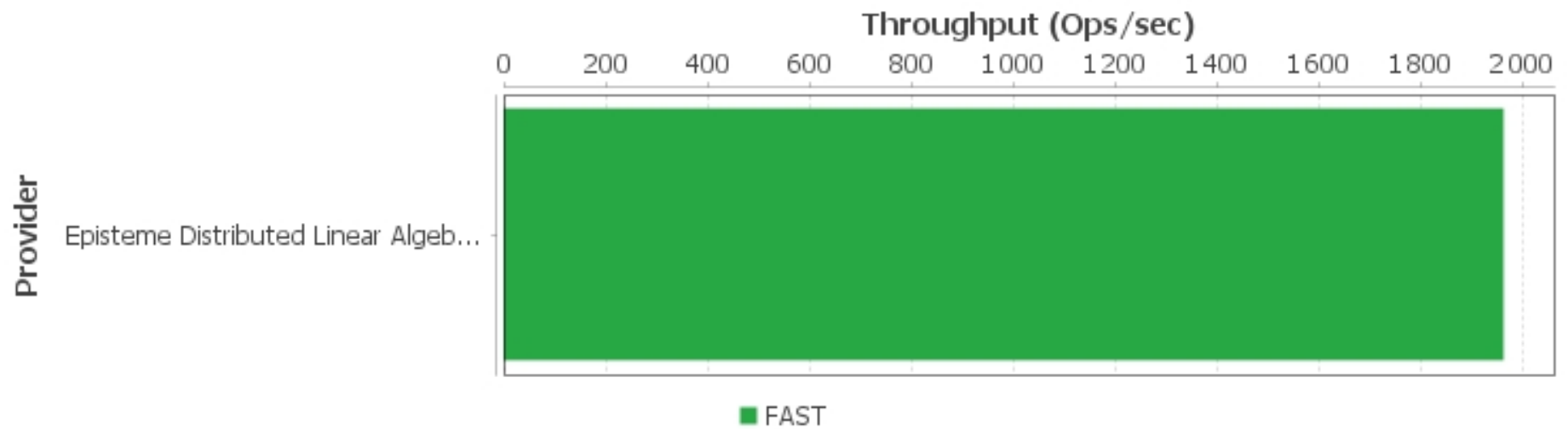
Operation Comparison: R:Func:Sqrt (FAST)

R:Func:Sqrt Performance Comparison



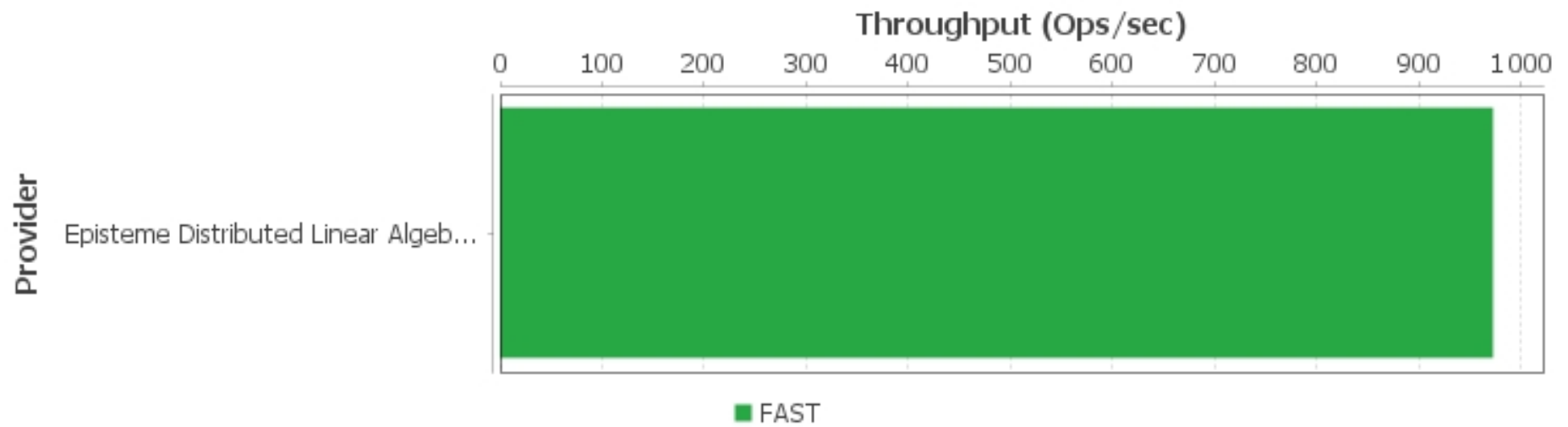
Operation Comparison: R:Func:Cbrt (FAST)

R:Func:Cbrt Performance Comparison



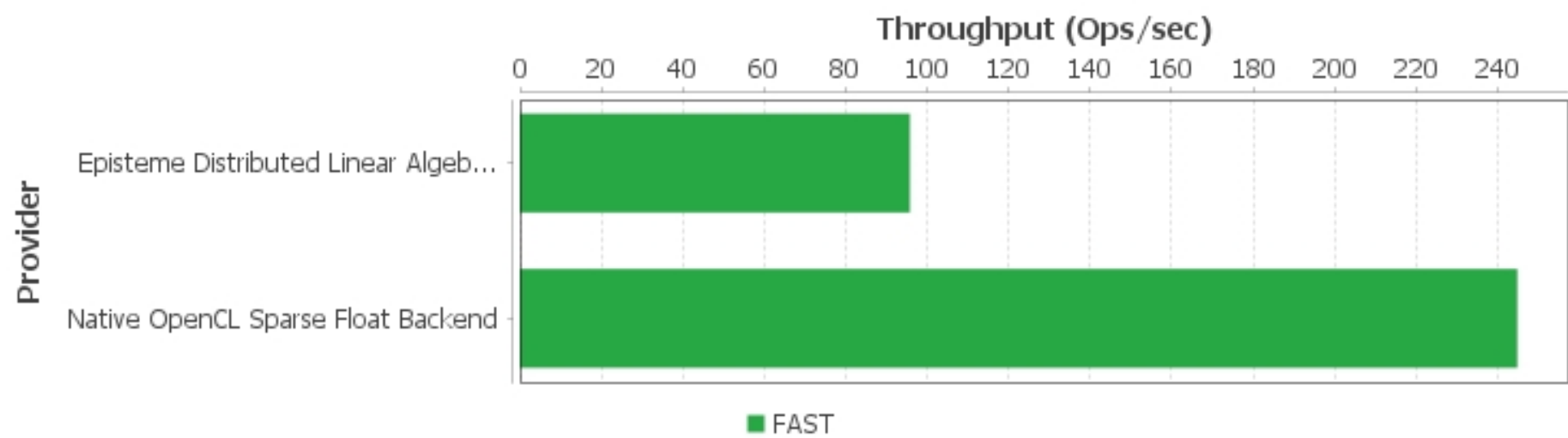
Operation Comparison: R:Func:Pow (FAST)

R:Func:Pow Performance Comparison



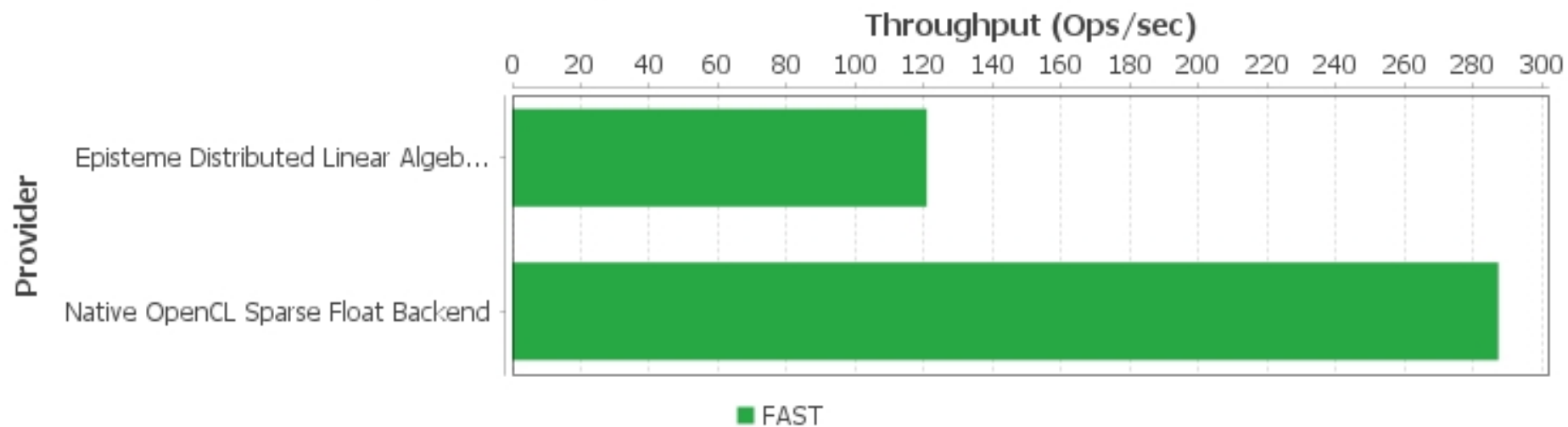
Operation Comparison: R:Sparse:BiCGSTAB (FAST)

R:Sparse:BiCGSTAB Performance Comparison



Operation Comparison: R:Sparse:ConjGrad (FAST)

R:Sparse:ConjGrad Performance Comparison



Operation Comparison: R:Sparse:GMRES (FAST)

R:Sparse:GMRES Performance Comparison

